Michigan Tech

Poincaré Sections

From a flow to a return map

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Outline

- **Flow**: advances states in time.
- **Trajectory** and **Orbit**: motion in time and its path.
- **Equilibrium Point**: a trajectory fixed at one state.
- **Poincaré Section**: a surface crossed by trajectories.
- **Return Map**: sends each crossing point to the next.
- **Periodic Orbit**: the closed path of a periodic trajectory.
- Rössler flow: plane and hilltop sections.

Poincaré and Qualitative Dynamics

- Poincaré studied trajectories without explicit solutions in his 1881–82 memoirs [1, 2].
- A trajectory can cross a chosen surface at distinct points.
- Those crossing points form a discrete sequence.

Flow

- An autonomous system has state $\boldsymbol{y} \in \mathbb{R}^n$:

$$\boldsymbol{y}' = \boldsymbol{f}(\boldsymbol{y}).$$

- $\varphi_t(\boldsymbol{y}_0)$ is the state at time t from $\boldsymbol{y}_0$:

$$\varphi_0(\boldsymbol{y}_0) = \boldsymbol{y}_0, \qquad \frac{d}{dt}\varphi_t(\boldsymbol{y}_0) = \boldsymbol{f}\big(\varphi_t(\boldsymbol{y}_0)\big).$$

- The family φ_t is the **Flow**. A **Trajectory** gives the state $\varphi_t(\boldsymbol{y}_0)$ as time passes. Its **Orbit** is the path traced in state space.
- An **Equilibrium Point** $\boldsymbol{q}_*$ satisfies $\boldsymbol{f}(\boldsymbol{q}_*) = \boldsymbol{0}$. Its trajectory is constant and its orbit is the single point $\{\boldsymbol{q}_*\}$.

Poincaré Sections

- A **Poincaré Section** Σ is a surface of dimension $n-1$ crossed by trajectories:

$$\Sigma = \{\boldsymbol{y} : h(\boldsymbol{y}) = 0\}, \qquad \nabla h \neq 0.$$

- A **Transverse Crossing** at $\boldsymbol{q} \in \Sigma$ satisfies

$$\nabla h(\boldsymbol{q}) \cdot \boldsymbol{f}(\boldsymbol{q}) \neq 0.$$

- Keep crossing points in one direction, such as $\nabla h(\boldsymbol{q}) \cdot \boldsymbol{f}(\boldsymbol{q}) > 0$. We use Σ for this directed section in the return map. In three dimensions, the surface is two-dimensional.

First Return Map

- Poincaré (1881) contrasted the "geometric study of the curve" with the "numerical calculation of its values" (our translations) [1].
- In his 1890 three-body study, successive crossing points on a transverse section revealed how trajectories return [3].

1. Start at a crossing point $\boldsymbol{q} \in \Sigma$. Follow its trajectory to the next crossing in the same direction.
2. The **First Return Time**, when it exists, is
$$\tau(\boldsymbol{q}) = \min\{t > 0 : \varphi_t(\boldsymbol{q}) \in \Sigma\}.$$
3. The **First Return Map** sends one crossing point to the next:
$$P(\boldsymbol{q}) = \varphi_{\tau(\boldsymbol{q})}(\boldsymbol{q}), \qquad P : D \subseteq \Sigma \longrightarrow \Sigma.$$
D contains returning crossing points. Successive points satisfy $\boldsymbol{q}_{k+1} = P(\boldsymbol{q}_k)$.

Periodic Orbits

- A **Periodic Solution** satisfies, for some $T > 0$,

$$\boldsymbol{y}(t + T) = \boldsymbol{y}(t) \qquad \text{for every } t.$$

- For a nonconstant periodic solution, the smallest such T is the **Minimal Period**. Its **Periodic Orbit** is the closed path:

$$\Gamma = \{\varphi_t(\boldsymbol{q}) : 0 \leq t < T\}.$$

- An equilibrium trajectory is constant. A periodic trajectory traces a closed orbit.

Periodic Points of Return Map

- A **Fixed Point** is a crossing point with $P(\boldsymbol{q}) = \boldsymbol{q}$. The flow trajectory closes after one return at time $T = \tau(\boldsymbol{q})$.
- A **Periodic Point** of minimal map period m satisfies

$$P^m(\boldsymbol{q}) = \boldsymbol{q}, \qquad P^j(\boldsymbol{q}) \neq \boldsymbol{q} \quad (1 \leq j < m).$$

- The flow trajectory closes after m crossings. Its elapsed time is

$$T = \sum_{j=0}^{m-1} \tau\big(P^j(\boldsymbol{q})\big).$$

- m counts crossings. T measures elapsed time.

Rössler Flow

- For $\boldsymbol{y} = (y_1, y_2, y_3)^T$, use

$$\boldsymbol{y}' = \boldsymbol{f}(\boldsymbol{y}) = \begin{pmatrix} -y_2 - y_3 \\ y_1 + a y_2 \\ b + y_3(y_1 - c) \end{pmatrix}, \qquad a = b = 0.2, \quad c = 5.7.$$

- The first two coordinates rotate. The third lifts and folds each trajectory.
- Typical attractor trajectories do not repeat exactly [4].

Rössler Flow in Words

- When $y_3 \approx 0$, the first two equations give an outward spiral in the (y_1, y_2) plane.
- When $y_1 > c$ and $y_3 > 0$, y_3 grows. The $-y_3$ term then pushes y_1 below c.
- Rössler described the return map this way:

  "The flow in state space allows for a 'folded' Poincaré map (horseshoe map)."

  [4]

Plane Section

1. Choose the plane $y_1 = 0$. Keep crossing points with $y_1' > 0$:

$$\Sigma = \{\boldsymbol{y} : y_1 = 0,\ -y_2 - y_3 > 0\}.$$

2. Check the crossing direction using $h(\boldsymbol{y}) = y_1$:

$$\nabla h \cdot \boldsymbol{f} = -y_2 - y_3 > 0.$$

3. Record (y_2, y_3) at each crossing point:

$$(y_{2,k}, y_{3,k}) \longmapsto (y_{2,k+1}, y_{3,k+1}).$$

- Each crossing point has two section coordinates, even when points lie close to a curve.

Hilltop Section

1. A hilltop is a nondegenerate local maximum of y_3:

$$y_3' = 0, \qquad y_3'' < 0.$$

2. Set $h(\boldsymbol{y}) = f_3(\boldsymbol{y}) = b + y_3(y_1 - c)$. Along the flow,

$$\frac{d}{dt} h\big(\boldsymbol{y}(t)\big) = y_3(-y_2 - y_3) + (y_1 - c)h(\boldsymbol{y}).$$

3. The **Hilltop Section** retains downward crossing points on $h = 0$:

$$\Sigma_{\text{top}} = \{\boldsymbol{y} : b + y_3(y_1 - c) = 0,\ y_3(-y_2 - y_3) < 0\}.$$

- This section is curved rather than horizontal.

Hilltop Return Map

- Record the full crossing state $\boldsymbol{q}_k$ at each selected maximum:

$$\boldsymbol{q}_{k+1} = P_{\text{top}}(\boldsymbol{q}_k).$$

- Since $b = 0.2$, the section equation gives $y_3 \neq 0$ and

$$y_1 = c - \frac{b}{y_3}.$$

  Thus (y_2, y_3) are section coordinates.
- Peak height alone need not determine the next peak. The map uses both coordinates and only points that return.

Summary

- A trajectory gives the state over time. Its orbit is the path traced [1, 2].
- A Poincaré Section is a surface. Trajectories cross it at points [1, 2].
- The Return Map sends each crossing point to the next [1, 2].
- A periodic trajectory traces a closed orbit. Periodic points of the Return Map identify it [1, 2].
- The Rössler flow admits both a plane section and a hilltop section [4].

References

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[4] Otto E. Rössler.
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